INF10025 Data Management and Analytics

Task 5 – Pass Submission

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A. Task Pass5a

In Task Pass5a, students are provided with a scenario describing Pete's Pools, a business that operates a number of swimming centres. Specifically, each centre has a name, an establishment date and a description, while clients, who have an ID and a name, can join one or more of these centres, and the date on which each client joined is recorded. Based on this case study, students are required to draw an Entity Relationship Diagram (ERD) in which all many-to-many relationships are fully expanded into many-to-one relationships and no surrogate keys are introduced. Furthermore, the ERD may be neatly hand drawn or created in a package such as Visio or draw.io, but its symbols must conform to those used throughout the ERD lectures this semester. Consequently, the diagram must clearly indicate the strong and weak entities, all identifiers and all relationship names.

> ***Pete's Pools*** *have a number of swimming centres. Each centre has a name, establishment date, and a description. Clients (who have an ID and name) can join one or more of these swimming centres. As a client joins a centre, the date that the client joined is to be recorded.*

Figure 1: Case Study Narrative of Pere's Pools for Task Pass5a.

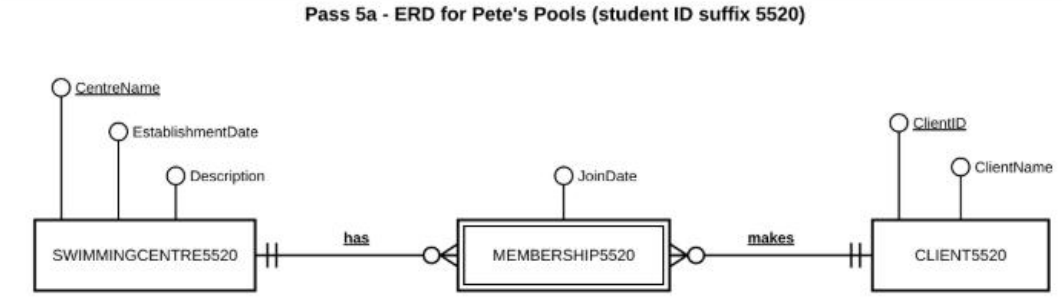


Figure 2: Entity Relationship Diagram (ERD) for Pete's Pools Swimming Centres.

B. Task Pass5b

In Pass 5b, students are required to create a relational schema based on the ERD that was produced in Pass 5a, clearly indicating all primary and foreign keys.

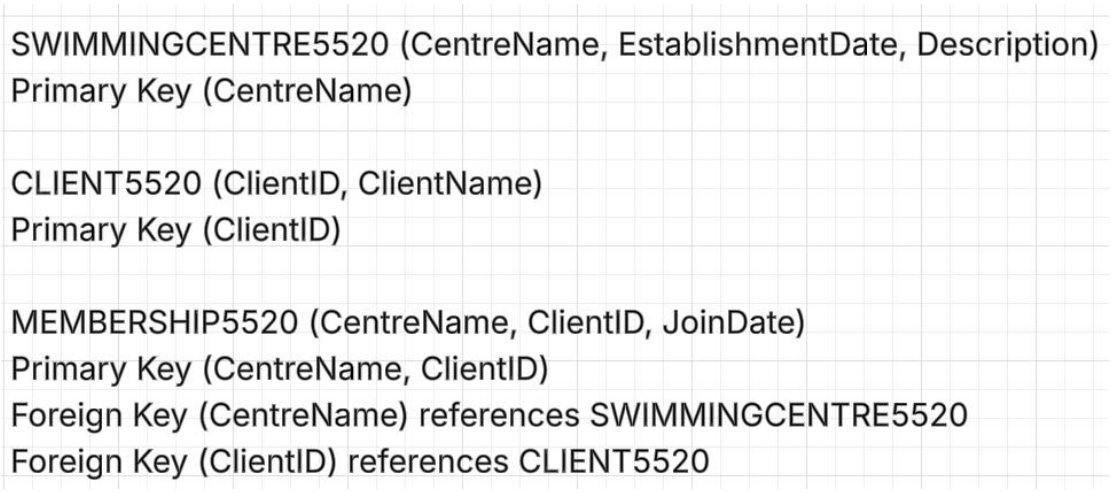
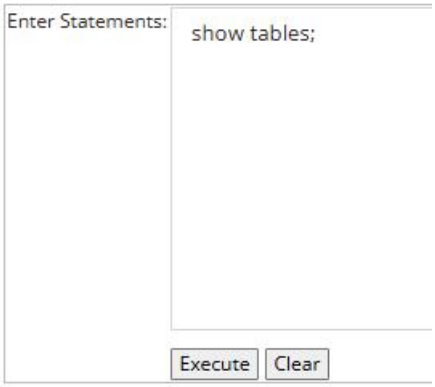


Figure 3: Relational Schema for Pete's Pools Swimming Centres.

C. Task Pass5c



SELECT TABLE_NAME
FROM ALL_TABLES
WHERE OWNER =
(SELECT USER FROM
DUAL)

TABLE_NAME
ACTOR5520
ASSIGNMENT5520
ATTENDEE5520
CAMPAIGN5520
CAR5520
CASTING5520
CLIENT5520
COLOURTYPE5520
COMEDIAN5520
CUSTOMER5520
DRIVER5520
JOB5520
MOVIE5520
PLAYER5520
RATING5520
RATINGSHOW5520
SHOW5520
SWIMMINGCENTRE5520

OK. 18 rows selected.

Figure 4: Checking Existing Tables in iSQL Junior Before Creating the Pete's Pools Tables.

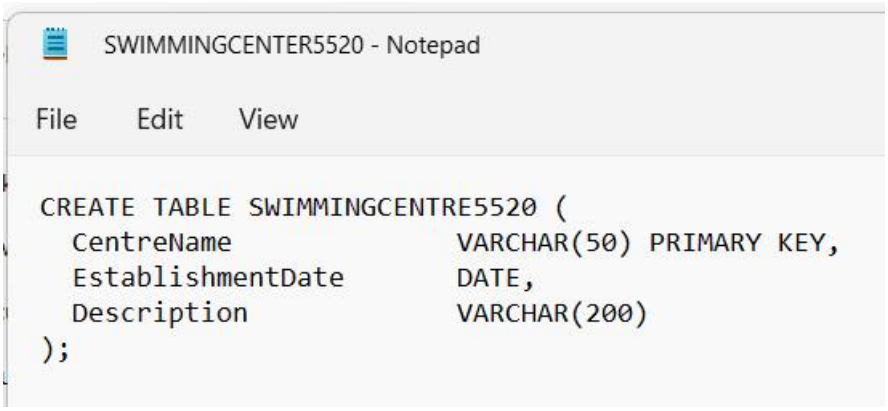
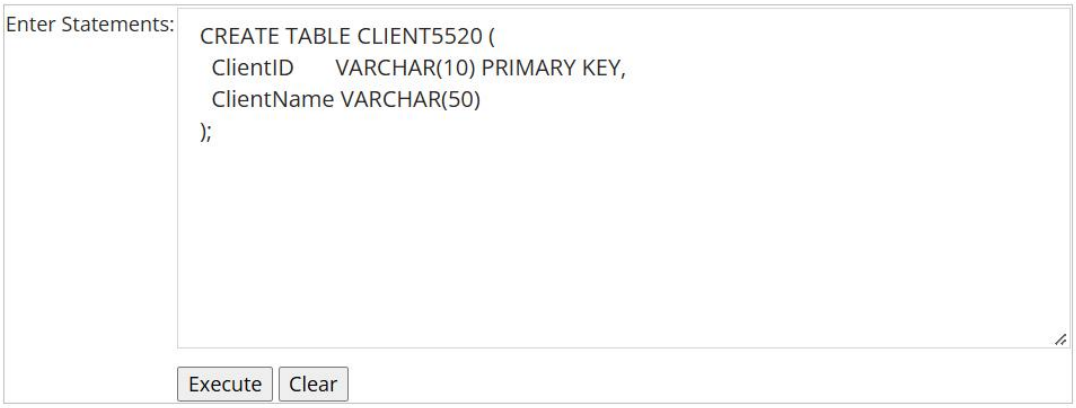
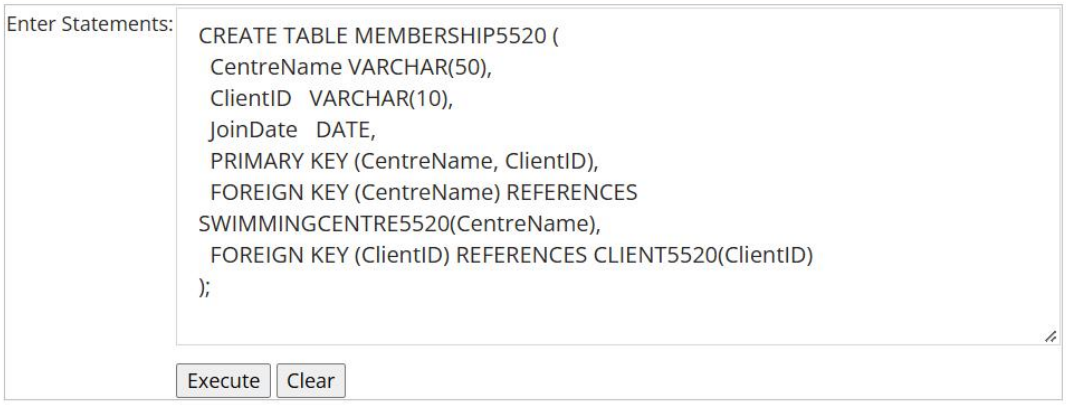


Figure 5: SQL Statement for Creating SWIMMINGCENTRE5520 Table.



OK. 0 rows

Figure 6: SQL Statement and Execution Result for Creating CLIENT5520 Table.



OK. 0 rows

Figure 7: SQL Statement and Execution Result for Creating MEMBERSHIP5520 Table.

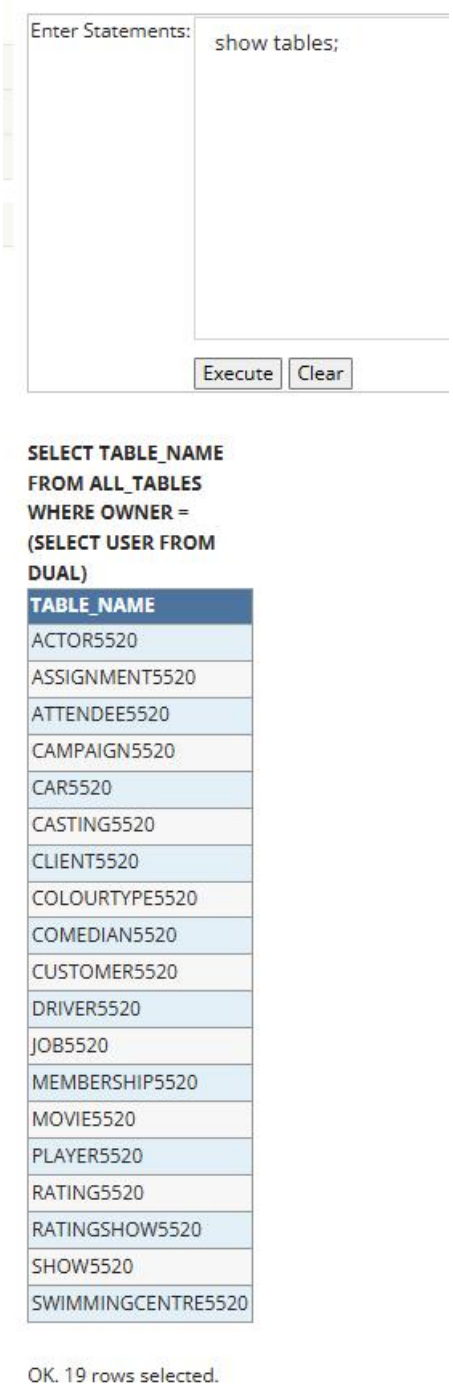


Figure 8: Verification of Created Tables in iSQL Junior.

D. Task Pass5d

Normalization is the systematic process of organizing the columns and tables of a relational database to minimize data redundancy (duplication) and improve data integrity.

The primary goals and key aspects of normalization include:

- Eliminating Redundancy: Ensuring that each piece of data is stored in only one place, which saves storage space and prevents duplicate entries.
- Preventing Data Anomalies: Protecting the database against logical errors during data operations, specifically:
    - *Insertion anomalies* (inability to add data without prior unrelated data).
    - *Update anomalies* (inconsistencies caused by failing to update all duplicate records).
    - *Deletion anomalies* (accidental loss of valid data when a related record is deleted).
- Progressive Normal Forms (NFs): The process is typically carried out through a series of formal stages called Normal Forms—moving from First Normal Form (1NF) (ensuring atomic, indivisible values and primary keys) to Second (2NF) and Third Normal Forms (3NF) (removing partial and transitive dependencies) to ensure tables are logically structured and properly related via Foreign Keys.

E. Task Pass5e

CustId	Name	Phone	CarRego	MakeModel	StartDate	EndDate
125	John Coles	0401112233	1AU8HK	Mazda 3	31/08/2020	7/09/2020
125	John Coles	0401112233	1LM3AB	Huyndai i30	14/11/2020	21/11/2020
278	Erin Trump	0466121455	1AU8HK	Mazda 3	12/09/2020	19/08/2020
278	Erin Trump	0466121455	1KA2CA	Toyota Camry	1/10/2020	8/10/2020
278	Erin Trump	0466121455	1CZ8JK	Mazda 3	10/11/2020	12/11/2020
278	Erin Trump	0466121455	1AU8HK	Mazda 3	26/11/2020	1/12/2020
721	Lynda Rodgers	0423544117	1LM3AB	Huyndai i30	10/09/2020	13/09/2020

Conclusion

From my perspective, Pass 5b provided the relational schema derived from the Pass 5a ERD, which turns the conceptual design into a structure that can be built as real tables. To begin with, the schema consists of three relations: SWIMMINGCENTRE5520(CentreName, EstablishmentDate, Description), CLIENT5520(ClientID, ClientName) and MEMBERSHIP5520(CentreName, ClientID, JoinDate). In MEMBERSHIP5520, the foreign keys CentreName and ClientID reference the two parent relations.

Through this task, I learnt how to convert an ERD into a schema. Firstly, the many-to-many relationship is resolved by the weak entity MEMBERSHIP5520, whose primary key is formed from the borrowed identifiers of both parents. Furthermore, I learnt that foreign keys are placed on the “many” side and must match the primary keys they reference, so that referential integrity is maintained. As a result, this task prepared me well for writing the DDL in Pass 5c. Overall, it also helped me understand that a correct schema reduces redundancy and prevents data anomalies.